ASSIGNMENT–02 | GENERATIVE AI

Comparative Study and Practical Analysis of Generative AI Models

Autoencoder • Variational Autoencoder • DCGAN • Stable Diffusion CO3 & CO4

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Branch / Batch	CSE-AIML / A2	Course	Generative AI

1. Objective

This report compares four generative models: **Autoencoder (AE)** and **Variational Autoencoder (VAE)** for image reconstruction (CO3), and **Deep Convolutional GAN (DCGAN)** and a **pretrained Diffusion Model (Stable Diffusion v1.5)** for image generation (CO4). Each model is studied theoretically and then analysed through outputs from a Google Colab notebook.

2. Task 1: Comparative Study of Generative AI Models

Table 1. Comparison of AE, VAE, DCGAN and Stable Diffusion

Criteria	Autoencoder (AE)	Variational AE (VAE)	DCGAN	Pretrained Diffusion (Stable Diffusion v1.5)
Working principle	Encoder compresses an image into a fixed latent code; decoder rebuilds the image. Trained by minimising pixel reconstruction error (deterministic).	Encoder predicts a mean and log-variance, so each image maps to a **distribution**. A latent vector is sampled (reparameterisation trick) and decoded. Loss = reconstruction + KL divergence.	Two networks play a minimax game: a **generator** turns random noise into images, a **discriminator** judges real vs. fake. Each improves by competing.	Noise is added to an image step by step; a network learns to **reverse** this, denoising from pure noise to an image. In latent space, guided by text through classifier-free guidance.
Architecture & major components	3 Conv layers (3→32→64→128 channels, 32×32→4×4), bottleneck, 3 ConvTranspose layers, Sigmoid output. Loss: MSE.	Same Conv encoder + Flatten, two linear heads (μ, log σ^2; 128-D latent), sampling layer, linear + ConvTranspose decoder. Loss: BCE + KL.	Generator: z (100-D) → 4 ConvTranspose layers with BatchNorm, ReLU, Tanh. Discriminator: 4 strided Conv layers, BatchNorm, LeakyReLU, Sigmoid. Loss: BCE.	CLIP text encoder, U-Net denoiser with cross-attention, VAE (image ↔ latent) and a noise scheduler. 25 denoising steps, guidance scale 7.5.
Advantages	Simple and fast to train; sharp reconstructions when the bottleneck is wide; learns useful compact features.	Smooth, structured latent space; supports sampling, interpolation and generation; principled probabilistic framework.	Sharper, more textured outputs than VAE; fast one-pass sampling; no explicit likelihood needed.	State-of-the-art fidelity and diversity; stable training; flexible text control over content and style.
Limitations	Latent space is irregular, so it generates poorly; may simply copy input if the bottleneck is too large.	Blurry outputs; trade-off between reconstruction and KL term; sensitive to loss weighting.	Unstable training, mode collapse risk; needs long training; limited resolution; no text control.	Slow iterative sampling; heavy GPU and memory need; may ignore prompt details; bias and safety concerns.
Applications	Denoising, compression, anomaly detection, dimensionality reduction, feature learning.	Generative modelling, latent interpolation, data augmentation, anomaly detection, representation learning.	Synthetic data, texture and face synthesis, data augmentation, image-to-image translation (variants).	Text-to-image art, inpainting, image editing, super-resolution, design and advertising.
Observed output (our experiment)	**Closest to the original.** Colours and shapes preserved; mild blur on edges and fine texture.	**Heavily blurred.** Only dominant colours and coarse layout survive; objects are hard to identify.	**Textured but incoherent.** Realistic colours and contrast, but no clear objects after 10 epochs (32×32).	**Coherent and detailed.** A clear red vintage car with a stylised illustration look (512×512) that follows the prompt.

3. Task 2: Practical Implementation and Output Analysis

3.1 Experimental setup

Ready-made PyTorch implementations were run on Google Colab (GPU). AE, VAE and DCGAN used **CIFAR-10** (32×32 colour images, batch size 128, Adam, 10 epochs each); the same eight test images were fed to AE and VAE for a like-for-like comparison. Stable Diffusion used the pretrained runwayml/stable-diffusion-v1-5 pipeline with a text prompt.

3.2 Activity A: Image reconstruction (CO3)



Figure 1. *Input test images (top row), Autoencoder reconstructions (middle row) and VAE reconstructions (bottom row).*

Aspect	Autoencoder (AE)	Variational Autoencoder (VAE)
Latent size	128×4×4 = 2048 values (about 1.5× compression of 3072 input values)	128 values (about 24× compression)
Final training loss	MSE = 0.0022 (epoch 10)	BCE + KL = 1820.81 per image (epoch 10); not directly comparable to AE
Blur / distortion	Mild blur; slight colour softening on small objects	Severe blur; shapes melt into smooth colour patches
Loss of detail	Edges and small textures (e.g., ship rigging, text on the plane) soften	Almost all fine detail, texture and object identity lost

Analysis: The AE reproduces the input closely, with correct colours, shapes and object layout, and only a mild blur. The VAE output is far blurrier and mostly shows blobs of the right colours. Two reasons explain this: the VAE squeezes the image through a much smaller latent code (128 vs. 2048 values), and its KL term forces the latent space to stay smooth and close to a normal distribution. In return the VAE has a well-structured latent space that can generate new samples, which a plain AE does not offer.

3.3 Activity B: Image generation (CO4)



Figure 2. *Images generated by the DCGAN generator from random noise after 10 epochs on CIFAR-10 (8 of 16 fixed-noise samples).*



Figure 3. *Stable Diffusion v1.5 output for the prompt "A small red vintage car parked beside a tree, colorful illustration".*

DCGAN (trained 10 epochs, 32×32)

- **Realism:** realistic colours and textures, but no recognisable objects; shapes are fragmented and noisy.
- **Training:** final D loss 0.290, G loss 3.467; G loss dropped to 0.132 at epoch 8, a sign of unstable training.
- **Limitations:** mode-collapse risk, long training needed, no control over content.

Stable Diffusion v1.5 (pretrained, 512×512)

- **Realism & detail:** a recognisable car with wheels, bumper, windows and door, sharp edges and rich grass and foliage strokes.
- **Limitations:** the "small" car fills and is cropped by the frame, the tree appears only as foliage, and the door line shows a slight artifact; also needs a GPU and 25 slow denoising passes.

Analysis: After only 10 epochs the DCGAN produces plausible colours and textures, but the shapes are not recognisable objects, and its loss curve shows the instability typical of GAN training. Stable Diffusion, trained on a vast dataset, produces a coherent, detailed and prompt-driven image, at the cost of larger compute and slower sampling. In short, the AE beats the VAE on reconstruction fidelity, and diffusion beats the DCGAN on generation realism.

4. Task 3: Group Reflection and AI Usage Documentation

AI tools used	**Claude** for structuring the comparative table, drafting and refining the report text; **Google Colab** (GPU) with PyTorch, torchvision and Hugging Face Diffusers for the experiments; other assistants (ChatGPT, Gemini) may be listed here if used.
Important prompts	• "Compare Autoencoder, VAE, DCGAN and Stable Diffusion on working principle, architecture, advantages, limitations, applications and output quality in one table." • "Why is a VAE reconstruction blurrier than an Autoencoder reconstruction on CIFAR-10?" • "Explain mode collapse and how to recognise it in DCGAN samples." • "Analyse the outputs of my Colab notebook and write the reconstruction and generation analysis for the report."
Verification process	• Re-ran the notebook ourselves and checked every claim about output quality against the actual images and logged losses. • Cross-checked AI-generated theory with original papers and official documentation (see References). • Confirmed architecture details (layers, channels, latent size, loss functions) directly from the notebook code.
Key learning outcomes	• **Latent design drives quality:** the AE's wide bottleneck gives sharp reconstructions, while the VAE's compact, regularised latent space trades sharpness for smoothness and generative ability. • **Reconstruction is not generation:** AE/VAE learn to rebuild inputs; DCGAN and diffusion create new images from noise. • **Quality vs. cost:** diffusion gives the best realism but needs large pretrained models and slow sampling; DCGAN is light but unstable.
My contribution	ran the Colab notebook (AE, VAE, DCGAN, Stable Diffusion), captured the output screenshots and wrote the output analysis.

5. Conclusion

The AE gave the most faithful reconstructions, whereas the VAE traded sharpness for a smooth, generative latent space. For generation, the DCGAN was lightweight but unstable and low in detail, while the pretrained diffusion model delivered the most realistic and controllable results.

References

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